Fundamentals of AI – Unit 3

1 Define descriptive analytics and give one example of its business application.

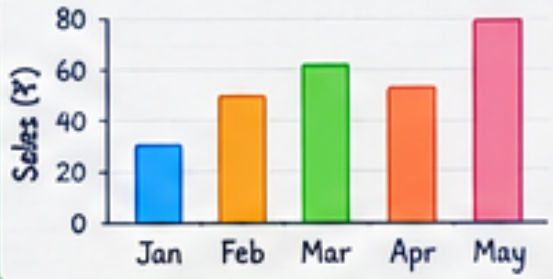
Descriptive Analytics

- Descriptive analytics is the analysis of historical data to understand what has happened in the past.
- It summarizes large amounts of data and converts it into meaningful information using reports, dashboards, charts and visualizations.
- It helps businesses identify trends, patterns and key insights for better decision making.

Key Characteristics

- Uses historical data
- Summarizes data
- Answers the question: What has happened?
- Uses charts, reports and dashboards.

Example: Monthly Sales Report (Retail)



Business Application

A retail company uses descriptive analytics to analyze monthly sales data and identify which sold the most in the past quarter.



2 What is predictive analytics? State one use case in the retail industry.

Predictive Analytics

- Predictive analytics uses historical data, statistical techniques and machine learning models to predict what is likely to happen in the future.
- It identifies patterns from past data and uses these patterns to forecast future outcomes.
- It helps businesses reduce risks and make proactive decisions.

Key Characteristics

- Uses historical and current data
- Applies statistical models / ML
- Answers the question: What is likely to happen?
- Provides probabilities and predictions

How Predictive Analytics Works?



Use Case (Retail Industry)

A retail company can use predictive analytics to forecast future product demand based on past sales data, seasonal trends and customer behaviour. This helps in optimizing inventory and avoiding stockouts.

3 What is prescriptive analytics? How does it differ from predictive analytics?

Prescriptive Analytics

- Prescriptive analytics recommends the best possible actions to achieve a desired outcome. It uses optimization techniques, and AI models and AI models to suggest decisions based on predicted results.
- It not only predicts what will happen but also suggests what should be done.

Example

- A retail company uses prescriptive analytics to recommend the optimal pricing strategy for a product to maximize profit based on demand forecast, competitor prices and market conditions.



Difference between Predictive and Prescriptive Analytics

Aspect	Predictive Analytics	Prescriptive Analytics
Purpose	Predicts what is likely to happen	Recommends what action to take
Output	Forecasts (e.g. sales will increase)	Actionable recommendations (e.g. increase stock, adjust price)
Techniques	Statistical models, Machine Learning	Optimization, Simulation, AI, Decision Models
Business Question	What is likely to happen?	What should we do?
Example	Predict next month's sales	Suggest best pricing or inventory level

4 Define KPI (Key Performance Indicator) and name two examples used to measure AI impact.

KPI (Key Performance Indicator)

- A KPI is a measurable value used to evaluate the performance and success of a business objective. [illegible] initiatives, measure [illegible] determine whether the desired outcomes are achieved.

Characteristics of a Good KPI

- ✅ Measurable
- ✅ Relevant to business goals
- ✅ Time-bound
- ✅ Easy to understand
- ✅ Aligned with strategic objectives.

Examples of KPIs for AI Impact



1. Productivity Increase

Measures how much AI has improved the output or efficiency of a process.



2. Customer Satisfaction Score (CSAT)

Measures improvement in customer experience due to AI solutions.

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5 What is meant by 'AI bias'? Give one real-world example.

AI Bias

- AI bias refers to systematic and unfair outcomes in AI models caused by biased data, flawed algorithms or human bias in design, development and deployment.
- It can lead to discriminatory or unequal treatment of individuals or groups.



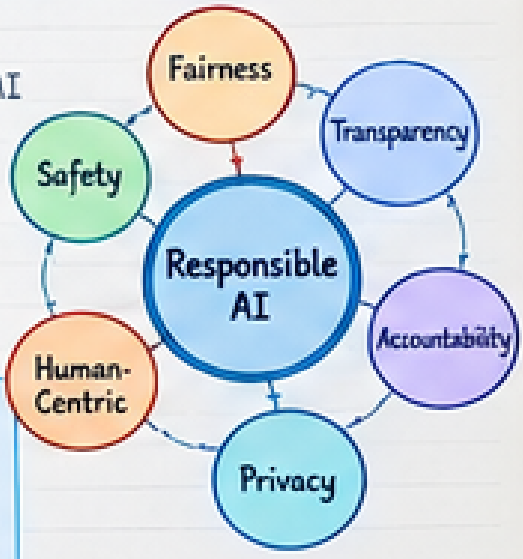
Real-world Example

- A recruitment AI tool (e.g. Amazon) showed lower recommendations for female candidates because it was trained on historical hiring data where most successful candidates were male.

6 Define 'responsible AI' and state two principles it is based on.

Responsible AI

- Responsible AI refers to the development and use of AI systems in a way that is ethical, fair, transparent and beneficial to society.
- It ensures that AI technologies are designed and deployed with consideration for human rights, fairness and long-term societal impact.



Two Key Principles

1. Fairness



Ensure non-discriminatory and unbiased outcomes for all individuals.

2. Transparency



Ensure clear explanation of how AI decisions are made.

7 What is data privacy? Why is it a concern in AI-driven business systems?

Data Privacy

- Data privacy refers to the protection of personal and sensitive information from unauthorized access, use or disclosure.
- In AI-driven business systems, large amounts of personal data (e.g. customer data, behaviour data) are collected and analyzed, which increases the risk of privacy breaches.

Why it is a concern?

- Risk of unauthorized access and data leaks
- Misuse of personal data
- Loss of customer trust
- Legal and regulatory issues (e.g. GDPR)
- Can lead to financial penalties



Fundamentals of AI – Unit 3

8 What is generative AI? Name one popular generative AI tool.

Generative AI

- Generative AI refers to a type of AI that can create new content such as text, images, videos, code or music by learning patterns from large amounts of data.
- It uses deep learning models, particularly large language models (LLMs) and generative models, to generate original and realistic content.

Common Applications

- Text generation (articles, reports)
- Image generation (art, design)
- Code generation (programming)
- Video and music generation

Popular Generative AI Tools



9 What is data governance? State its importance in AI implementation.

Data Governance

- Data governance is the framework of policies, processes, standards and practices to ensure that the information is consistent, accessible and used responsibly across an organization.
- It defines who can access data, how data is managed and how data quality is maintained.

Data Governance Framework



Importance in AI Implementation

- ✅ Ensures high-quality and reliable data for training AI models.
- ✅ Helps in maintaining data security and privacy.
- ✅ Improves data consistency and reduces errors.
- ✅ Supports regulatory compliance (e.g. GDPR, data protection laws).
- ✅ Builds trust in AI systems and enables ethical use of data.

Fundamentals of AI – Unit 3

10 Differentiate between descriptive, predictive, and prescriptive analytics with examples from at least two industries.

Aspect	Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
Purpose	Understand what has happened	Predict what is likely to happen	Recommend what should be done
Data Used	Historical data	Historical + current data	Historical + current data + business rules
Techniques	Data aggregation, visualization	Statistical models, Machine Learning	Optimization, Simulation, AI
Key Question	What has happened?	What is likely to happen?	What should we do?
Output	Reports, dashboards	Forecasts, predictions	Actionable recommendations
Example (Retail)	Monthly sales report	Predict next month's demand	Recommend optimal inventory and pricing
Example (Manufacturing)	Production summary report	Predict machine failure	Suggest maintenance schedule and resource allocation

Industry Examples

Retail Industry

- Descriptive : Total sales by product category
- Predictive : Forecast customer demand
- Prescriptive : Recommend optimal pricing and inventory

Manufacturing Industry

- Descriptive : Production output and defect rates
- Predictive : Predict equipment failure
- Prescriptive : Suggest maintenance schedule and resource allocation

11 How does AI-driven decision-making improve business strategy? Illustrate with a case example.

Impact on Business Strategy

- AI-driven decision-making provides data-driven insights, helps identify new opportunities, reduces risks, and enhances strategic decision-making. Suggest
- It allows businesses to better understand customers, optimize operations and create competitive advantages.

Benefits

- ✅ Improves accuracy of decisions
- ✅ Identifies market opportunities
- ✅ Enhances customer experience
- ✅ Optimizes operational efficiency
- ✅ Supports long-term strategic planning

Case Example : Netflix

Netflix uses AI to analyze user behaviour and viewing patterns personalized content. This has improved customer retention, increased engagement and helped Netflix maintain a competitive advantage in the streaming industry.

12 Explain how organizations use KPIs to measure the business impact of AI adoption. Give examples from manufacturing and retail.

How Organizations Use KPIs

- Organizations use KPIs to measure the success of AI adoption by tracking improvements in key business areas such as productivity, efficiency, cost reduction, revenue growth and customer satisfaction.
- KPIs help in evaluating whether AI initiatives are delivering the expected value and support data-driven decision making.

Manufacturing Industry – Examples

- Production efficiency (units/hour)
- Equipment downtime reduction (%)
- Defect rate reduction (%)
- Cost savings (₹)

Retail Industry – Examples

- Sales growth (%)
- Customer satisfaction score (CSAT)
- Increase in conversion rate (%)
- Inventory holding cost reduction (%)